

## XIAMEN, China

WMO: 591340

Lat: 24.4860N

Lon: 118.0798E

Elev: 456

StdP: 14.46

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 44.4       | 46.4 | 24.8                          | 19.2 | 53.2 | 29.5 | 24.0 | 53.0 | 20.8                  | 53.8 | 19.1 | 53.8 | 8.0                   | 70   | 0.371 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.2                   | 94.9            | 79.3 | 93.2 | 79.5 | 91.5 | 79.2 | 82.4                | 88.5 | 81.6 | 87.6 | 80.9 | 86.9 | 9.6                  | 200  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 80.9                            | 163.1 | 85.2 | 80.4 | 160.7 | 84.8 | 79.2 | 153.9 | 84.2 | 46.9          | 89.1 | 45.9 | 88.1 | 45.2 | 87.1 | 87.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 19.4          | 17.2 | 15.3 | 40.6                       | 99.2 | 2.9                | 1.9 | 38.5                                               | 100.6 | 36.8       | 101.7 | 35.2       | 102.8 | 33.0       | 104.2 |    |    |
| (5)               |      |      | 34.4                       | 83.9 | 3.3                | 1.1 | 32.0                                               | 84.7  | 30.0       | 85.4  | 28.2       | 86.0  | 25.8       | 86.8  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 72.1   | 57.6  | 58.0  | 61.9  | 69.6  | 76.2  | 81.5  | 85.1  | 84.6  | 82.3  | 76.7  | 69.6  | 61.3  |  |  |
|                                                                        | DBStd       | 11.02  | 5.17  | 6.03  | 6.14  | 5.90  | 4.64  | 3.84  | 2.69  | 2.99  | 3.37  | 3.98  | 4.71  | 5.17  |  |  |
|                                                                        | HDD50       | 13     | 5     | 6     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     |  |  |
|                                                                        | HDD65       | 751    | 236   | 205   | 136   | 24    | 1     | 0     | 0     | 0     | 0     | 0     | 13    | 136   |  |  |
|                                                                        | CDD50       | 8079   | 239   | 231   | 369   | 587   | 812   | 945   | 1087  | 1072  | 969   | 827   | 589   | 352   |  |  |
|                                                                        | CDD65       | 3342   | 5     | 10    | 39    | 162   | 348   | 495   | 622   | 607   | 519   | 362   | 152   | 21    |  |  |
|                                                                        | CDH74       | 31792  | 13    | 37    | 160   | 765   | 2367  | 5063  | 7705  | 7289  | 5457  | 2400  | 501   | 35    |  |  |
|                                                                        | CDH80       | 11968  | 0     | 3     | 24    | 174   | 642   | 1765  | 3500  | 3180  | 2033  | 580   | 66    | 1     |  |  |
| (14) Wind                                                              | WSAvg       | 7.0    | 7.1   | 6.9   | 6.6   | 6.3   | 6.1   | 6.6   | 6.7   | 6.4   | 7.3   | 8.6   | 7.9   | 7.6   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 46.7   | 1.4   | 2.5   | 4.0   | 5.7   | 6.1   | 7.7   | 5.4   | 5.8   | 4.6   | 1.1   | 1.4   | 0.9   |  |  |
|                                                                        | PrecMax     | 69.8   | 6.2   | 10.6  | 17.3  | 19.4  | 13.7  | 15.7  | 13.1  | 16.5  | 14.3  | 6.8   | 6.6   | 3.4   |  |  |
|                                                                        | PrecMin     | 33.1   | 0.0   | 0.1   | 0.4   | 1.2   | 0.9   | 1.5   | 0.4   | 0.1   | 0.2   | 0.0   | 0.0   | 0.0   |  |  |
|                                                                        | PrecStd     | 11.4   | 1.4   | 2.6   | 3.2   | 4.1   | 3.0   | 3.7   | 3.8   | 3.6   | 4.1   | 1.7   | 1.6   | 1.0   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 75.4  | 78.9  | 82.8  | 88.1  | 91.7  | 94.8  | 97.1  | 97.0  | 95.0  | 91.0  | 85.6  | 77.4  |  |  |
|                                                                        |             | MCWB   | 63.2  | 66.4  | 69.8  | 73.6  | 79.4  | 80.0  | 79.8  | 79.4  | 77.9  | 75.6  | 72.1  | 66.5  |  |  |
|                                                                        | 2%          | DB     | 71.2  | 73.5  | 78.5  | 84.5  | 88.2  | 91.7  | 94.9  | 94.8  | 91.9  | 87.8  | 82.0  | 73.8  |  |  |
|                                                                        |             | MCWB   | 61.7  | 64.3  | 67.8  | 72.9  | 77.5  | 79.7  | 79.6  | 79.2  | 77.9  | 74.1  | 69.8  | 64.0  |  |  |
|                                                                        | 5%          | DB     | 67.8  | 69.6  | 74.8  | 81.5  | 86.0  | 89.8  | 93.1  | 93.0  | 90.0  | 85.6  | 79.1  | 71.4  |  |  |
|                                                                        |             | MCWB   | 59.1  | 61.4  | 65.2  | 71.8  | 76.3  | 79.4  | 79.8  | 79.6  | 77.4  | 72.9  | 68.2  | 62.4  |  |  |
|                                                                        | 10%         | DB     | 64.7  | 66.3  | 70.6  | 78.6  | 83.8  | 87.8  | 91.4  | 91.1  | 88.2  | 82.8  | 76.9  | 68.8  |  |  |
|                                                                        |             | MCWB   | 57.9  | 59.5  | 62.9  | 70.2  | 74.7  | 78.9  | 79.6  | 79.3  | 76.6  | 71.3  | 67.5  | 61.3  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 66.1  | 69.8  | 72.7  | 77.6  | 82.0  | 83.0  | 83.6  | 83.5  | 82.3  | 79.0  | 74.9  | 69.4  |  |  |
|                                                                        |             | MCDB   | 71.9  | 76.0  | 80.3  | 83.8  | 87.7  | 89.2  | 91.5  | 90.5  | 88.7  | 84.7  | 80.8  | 74.3  |  |  |
|                                                                        | 2%          | WB     | 63.5  | 66.0  | 69.7  | 75.3  | 80.1  | 81.8  | 82.3  | 82.1  | 80.9  | 77.4  | 72.9  | 67.0  |  |  |
|                                                                        |             | MCDB   | 68.3  | 70.4  | 75.6  | 81.0  | 85.4  | 87.0  | 88.9  | 88.0  | 86.9  | 83.3  | 77.6  | 71.2  |  |  |
|                                                                        | 5%          | WB     | 61.4  | 63.3  | 67.4  | 73.7  | 78.6  | 80.9  | 81.4  | 81.3  | 79.9  | 75.8  | 71.4  | 65.0  |  |  |
|                                                                        |             | MCDB   | 66.0  | 67.8  | 72.1  | 78.9  | 83.3  | 86.1  | 88.0  | 87.4  | 85.7  | 82.1  | 76.2  | 69.6  |  |  |
|                                                                        | 10%         | WB     | 59.5  | 60.9  | 65.0  | 72.0  | 77.4  | 80.1  | 80.7  | 80.5  | 79.0  | 74.3  | 69.7  | 63.0  |  |  |
|                                                                        |             | MCDB   | 63.7  | 65.3  | 69.3  | 76.8  | 81.5  | 85.1  | 87.1  | 86.7  | 84.6  | 80.8  | 75.0  | 67.2  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 12.8  | 13.0  | 13.6  | 13.9  | 12.8  | 11.7  | 13.2  | 12.9  | 12.9  | 13.0  | 13.0  | 13.1  |  |  |
|                                                                        |             | MCDBR  | 16.3  | 17.8  | 18.9  | 17.8  | 15.6  | 14.1  | 15.3  | 15.4  | 14.6  | 14.4  | 14.9  | 15.0  |  |  |
|                                                                        |             | MCWBR  | 8.7   | 9.9   | 10.2  | 8.9   | 7.3   | 5.6   | 5.8   | 6.0   | 6.3   | 7.0   | 7.4   | 8.4   |  |  |
|                                                                        | 5% WB       | MCDBR  | 14.4  | 16.6  | 17.7  | 16.4  | 14.4  | 12.7  | 13.9  | 13.8  | 13.2  | 12.7  | 12.9  | 13.1  |  |  |
|                                                                        |             | MCWBR  | 8.6   | 10.5  | 11.0  | 8.9   | 7.3   | 5.7   | 5.9   | 6.1   | 6.1   | 6.5   | 7.1   | 8.1   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.543 | 0.608 | 0.706 | 0.706 | 0.594 | 0.542 | 0.517 | 0.573 | 0.591 | 0.575 | 0.534 | 0.522 |  |  |
|                                                                        | taud        |        | 1.848 | 1.744 | 1.584 | 1.635 | 1.913 | 2.065 | 2.140 | 1.962 | 1.891 | 1.864 | 1.939 | 1.919 |  |  |
|                                                                        | Ebn at Noon |        | 222   | 217   | 204   | 209   | 233   | 244   | 250   | 237   | 228   | 223   | 223   | 221   |  |  |
|                                                                        | Edh at Noon |        | 59    | 69    | 85    | 83    | 62    | 53    | 49    | 59    | 62    | 61    | 53    | 53    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 928   | 963   | 1106  | 1247  | 1368  | 1498  | 1827  | 1651  | 1506  | 1302  | 1020  | 930   |  |  |
|                                                                        | RadStd      |        | 136   | 167   | 146   | 192   | 209   | 172   | 133   | 152   | 132   | 157   | 108   | 118   |  |  |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           |       | +0.83   | N/A    | N/A     | +0.63 | +0.59 | N/A         | N/A   | +218  | +202  |
| (47) Regional (8 neighbors) |       | +0.65   | N/A    | N/A     | +0.95 | +0.32 | +0.40       | N/A   | N/A   | +245  |

Nomenclature: See separate page